

Chapter Eight Debugging and Optimization (Fixing Common Mistakes)

8-1: Why Didn't the Prompt Work? (Diagnosing 5 Common Mistakes)

Many people, when they don't get a good answer, say "AI is stupid!" But in 90% of cases, the problem is with the prompt itself. Let's get to know the 5 silent killers of prompts:

Mistake	Explanation	Example	Solution
1. Vague and general prompt	The model doesn't know exactly what you want	"Explain marketing"	Specify: "Write 5 content marketing strategies for an online clothing store"
2. Missing role	Answers become general and neutral	"Write a workout plan"	Say: "As a bodybuilding coach who has worked with beginners, write a plan"
3. Ignoring constraints	The model either writes too much or too little	"Summarize" (the model writes 500 words!)	Say: "Summarize in 100 words"
4. Wrong information in context	You send the model down the wrong path	If you say "My audience is elementary school kids" but use university-level terms, the model gets confused	Give context information honestly and accurately
5. Multitasking	The model doesn't know which task to prioritize	"Write an article, also give its code, and draw a table too"	Number the tasks or ask them in separate stages

Quick error diagnosis solution: If you got an answer and weren't satisfied, ask yourself: "Which of these 5 mistakes did I make?" Then fix that one part and send it again.

8-2: Iterative Refinement Technique (Instead of Rewriting, Tell the Model: "This Is Good, But Fix Such-and-Such Part")

The biggest mistake beginners make: When the first answer isn't satisfactory, they rewrite the entire prompt from scratch and send it again. This means starting over from zero!

The professional solution: Iterative Refinement means telling the model: "The output was good, but fix these 3 points." This way the model keeps the previous output as a base and only changes the parts you specified.

Practical example:

- **First prompt:** *"Write a sales email to introduce a Python programming course."*
- **Model's response:** A dry and administrative email.
- **Refinement prompt (without rewriting):** *"The email you wrote was good, but fix these 3 things: 1) Make the tone more intimate (like an old friend). 2) Add a line that addresses beginners' common fears. 3) End the email with a special offer (10% discount)."*

Golden tip: Use keywords like "Keep the same previous output, but...", "It was good, but...", "Great! Now add this part too..."

8-3: Escaping Hallucinations — How to Stop AI from Lying

Hallucination means AI willingly and with complete confidence delivers information that is **completely wrong or fabricated by its own mind**. For example, if you ask *"Who was the president of Iran in 1850?"* (when no such thing existed), the model happily delivers a fake name!

Why does this happen? Because the model is a "word prediction machine," and if it doesn't find an answer in its data, instead of saying "I don't know," it combines the most probable words to make an acceptable sentence!

3 solutions to escape hallucination:

1. Requesting sources and citations:

"Based on what source are you saying this information? Please also mention credible sources (with links or book names)."

2. Limiting to provided information:

"Answer only based on the information I give you in this conversation. Do not use your own general knowledge to fill in the gaps."

3. Requesting cross-check:

"Now that you've given the answer, please review it again. Is there any part you're not sure about? If so, mark it with the phrase 'estimated'."

Best technique: For critical tasks (medical, legal, engineering), always say: *"If you're not sure about the answer, instead of guessing, say 'I don't know' or 'I need more information'."*

8-4: A/B Testing on Prompts (Write Two Prompts and Compare the Results)

This technique turns you from a "user" into a **"prompt data scientist."**

A/B testing means for a specific task, you write **two different versions** of the prompt (by changing one variable) and give both to the AI, then compare the outputs to see which version answers better.

How to do A/B testing?

Scenario	Prompt A	Prompt B	Variable changed
Article request	"Write an article about healthy nutrition"	"As a nutrition expert, write an article about healthy nutrition for working mothers"	Role
Code request	"Write a Python code to sort a list"	"Write a Python code to sort a list and choose the most time-efficient algorithm"	Constraint (performance)

How to compare?

1. Run both.
2. Put the outputs side by side.
3. Answer 3 questions:
 - Which one was closer to my goal?
 - Which one had higher quality?
 - Which one had fewer errors?
4. Choose the winning version and if it needs modification, apply iterative refinement (8-2) to that same version.

★ Combined Practical Project for Chapter Eight (Become a Debugging Champion!)

Now let's combine all 4 techniques in a real scenario:

Suppose you asked AI: *"Write a simple program for managing daily expenses."* But you got a very complex program with 300 lines of code that doesn't work properly at all and is full of logical errors!

Your feedback prompt (using the techniques of this chapter):

*"The program you wrote is too complex and doesn't work.

[Debugging] I know where the problem is: I asked you to write a **simple** program for managing expenses, but you wrote an advanced program with a database (this was **Mistake #1: Vagueness** that caused you to misunderstand).

[Iterative Refinement] Keep the same program, but apply these fixes:

1. Use only simple Python lists (no database).
2. Reduce the code to under 50 lines.
3. Handle user input errors.

[Escaping hallucination] If you're not sure about any part of the code (for example, in the date management section), mark it with the comment "# estimated" so I can check it myself.

[A/B Testing] After writing the revised version, give **two versions**: one with full explanations (for beginners) and one with just the code (for professional programmers). Let's see which one I prefer."*

Practical Assignment for Chapter Eight (The Last Exercise Before the Final Project):

Write a prompt in which you asked AI to "give 5 ideas for an online business," but you got 5 general and empty ideas (like "online store" and "network marketing").

Your assignment: Send a feedback (refinement) prompt to the model in which:

1. You diagnose which of the 5 common mistakes (8-1) was the problem.
 2. You use the iterative refinement technique (8-2) and say exactly what to fix.
 3. You add a request to prevent hallucination (8-3) (for example, ask the model to give a real example of a successful brand for each idea).
 4. Finally, say that you want two versions (A/B) of the answer: one with risky ideas and one with low-risk ideas.
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